Taoufik Rachdi

Test Bench and Validation Engineer

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SUMMARY

Test Bench and Validation Engineer with 3+ years of experience in the automotive and aeronautical sectors. Currently designing test benches and dedicated PCBs (KiCad) for aeronautical equipment, and converting Acceptance Test Procedures (ATP) into automated test sequences. Previously led diagnostic validation (Diag/DTC) of In-Vehicle Infotainment systems for the Renault Alliance, using Vector tools, DDT2000, Robot Framework and Codebeamer. Passionate about embedded systems, hardware design, and test quality.

PROFESSIONAL EXPERIENCE

SERMA International — Jul 2023 – Present, Ariana

Test Bench Engineer – Aeronautical Equipment — Sep 2026 – Present

Test Bench Design & PCB Development

- Design and build test benches for the acceptance testing and validation of aeronautical equipment.
- Define the full bench architecture: hardware interfaces, instrumentation, power supplies, signal conditioning, and communication links.
- Translate Acceptance Test Procedures (ATP) into automated test sequences, reducing manual intervention and making every run repeatable and traceable.
- Design dedicated interface PCBs in KiCad end to end: schematic capture, component selection, placement, routing, DRC, and manufacturing files.
- Apply sound layout practices for signal integrity, power distribution, grounding, and manufacturability.
- Bring up and debug new boards through electrical measurements and troubleshooting, up to full functional validation.
- Help standardize test procedures, bench equipment, and validation methods across projects.
- Maintain technical documentation: schematics, PCB files, test procedures, validation reports, bench manuals.
- **Technical environment:** KiCad, ATP, bench instrumentation, signal conditioning, PCB bring-up.

Lead – Diagnostic (Diag/DTC) Function & PVS – Digital Avatar Function — Jul 2024 – Sep 2026

- Client: Renault Alliance – A IVI 2 Project (In-Vehicle Infotainment).
- Owned end-to-end validation of the Diag/DTC and Digital Avatar functions, from requirements to final status.
- Built the test strategy: planning, validation scope, risk identification, and requirement traceability.
- Authored and reviewed validation plans (PVAL), ensuring full functional coverage of both functions.
- Developed and maintained automated test suites with Robot Framework and Selenium to make campaigns faster and more repeatable.
- Ran test campaigns in Codebeamer and reported deviations with clear root-cause analysis.
- Managed the full defect lifecycle in Jira: detection, reporting, prioritization, and follow-up to closure.
- Prepared test benches, harnesses, and wiring, and configured test environments (calibration files, system configurations, diagnostic setups).
- Analyzed CAN communication and diagnostics with CANalyzer, CANoe, and Wireshark to isolate failures.
- Acted as the validation point of contact for cross-functional teams (PFS, PDEs, developers) and presented status in customer meetings.
- Drove improvements to test processes and campaign efficiency within the project.
- **Technical environment:** Codebeamer, Vector CANalyzer, CANoe, DDT2000, UDS, Jira, CAPL, Wireshark, Gateway, Shell/Bash scripting.

Validation Engineer – Diag/DTC Function — Jul 2023 – Jul 2024

- Client: Renault-Nissan-Mitsubishi Alliance – A IVI 2 Project (In-Vehicle Infotainment).
- Executed functional and non-functional test campaigns in Silk Central.
- Designed and ran integration, regression, and validation tests on test benches.
- Set up and configured validation benches and test environments.
- Analyzed results, documented deviations from requirements, and tracked defects in Jira.
- Automated CAN tests with CAPL scripts and maintained XML diagnostic scripts in DDT2000.
- Helped structure and continuously improve the team's testing activities.
- **Technical environment:** Silk Central, Vector CANalyzer, DDT2000, Jira, CAPL, Wireshark, IVI, Gateway.

Final Year Project – Smart Ways Innovation Feb – Jun 2023, Sfax
Designed an IoT gas and fire detection system based on ESP32, with real-time alerts through a mobile app and a web platform. Designed the system PCB in Altium Designer.

EDUCATION

National Engineering School of Sfax (ENIS) 2020 – 2023, Sfax
Engineering Degree – Electrical Engineering
Specialization: Embedded Electronics and Instrumentation.

Higher Institute of Applied Sciences and Technology of Gabès (ISSAT) 2018 – 2020, Gabès
Preparatory Cycle – Physics and Chemistry.

TECHNICAL SKILLS

- **Test Benches & Hardware:** Test bench architecture, instrumentation, signal conditioning, ATP automation, PCB bring-up and debugging.
- **Testing:** Manual and automated testing, bench testing, test case design and execution, test management (Codebeamer, Jira, Xray).
- **Automation:** Selenium, Robot Framework, Playwright, CAPL scripting.
- **Tools:** Vector CANalyzer, CANoe, DDT2000, Android Studio.
- **Protocols & Buses:** CAN, LIN, UDS, UART, I2C.
- **Programming:** C/C++, Python, Java, JavaScript, Embedded C.
- **Web & IoT:** HTML, CSS, Web selectors (DOM, XPath, CSS), Angular, Laravel.
- **Electronics:** STM32, FPGA, ESP32, Raspberry Pi, PCB design (KiCad, Altium Designer).

CERTIFICATIONS

- **ISTQB® Certified Tester – Foundation Level (CTFL)** ID: 115412
- **Certification in Manual and Automated Testing – CSF Training Center**

ADDITIONAL TRAINING

- **Automotive Cybersecurity Training – Udemy** Topics covered: Embedded systems security, vulnerability analysis, secure communication (CAN, LIN), ISO 21434 standard, ECU protection best practices.
- **Certified Training in Manual and Automated Testing – CSF Training Center** Topics covered: Test case design and execution, test automation with Selenium, Robot Framework, and Playwright, test campaign management, defect reporting and tracking.
- **STM32 Training – CSF Training Center** Topics covered: Embedded application development on ARM architecture (STM32), peripheral configuration with CubeMX, STM32CubeIDE, C programming, interrupts, UART/SPI/I2C communication.
- **CAPL Programming Training – Udemy** Topics covered: CAPL scripting for test automation, CAN network simulation, communication analysis and validation, use of CANoe/CANalyzer tools.

EXTRACURRICULAR ACTIVITIES

Electrical Club ENIS (CEE) – Organizer of ENIS 4.0 robotics competition.
IEEE SSCS ENIS Student Branch Chapter – Active member.

LANGUAGES

- **Arabic** – Native
- **French** – Professional proficiency
- **English** – Upper-intermediate
- **Spanish** – Beginner